Motion in a straight line

Mon May 04 2026

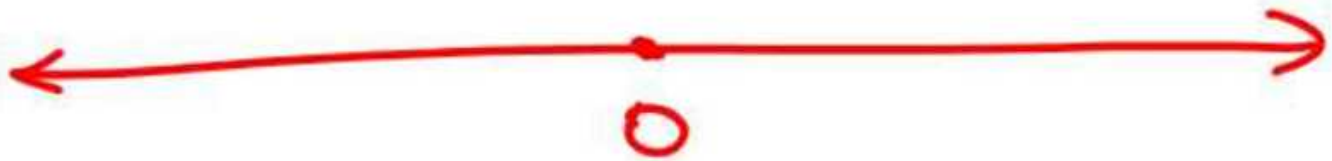


Motion in a straight line (1-D Motion)

Position (x)

$\vec{r}$ = Position vector

Location of an object wrt origin is called position of object. Position is a vector quantity



Displacement (s)

- Change in position is called as displacement
- Vector quantity

$s = x_2 - x_1$

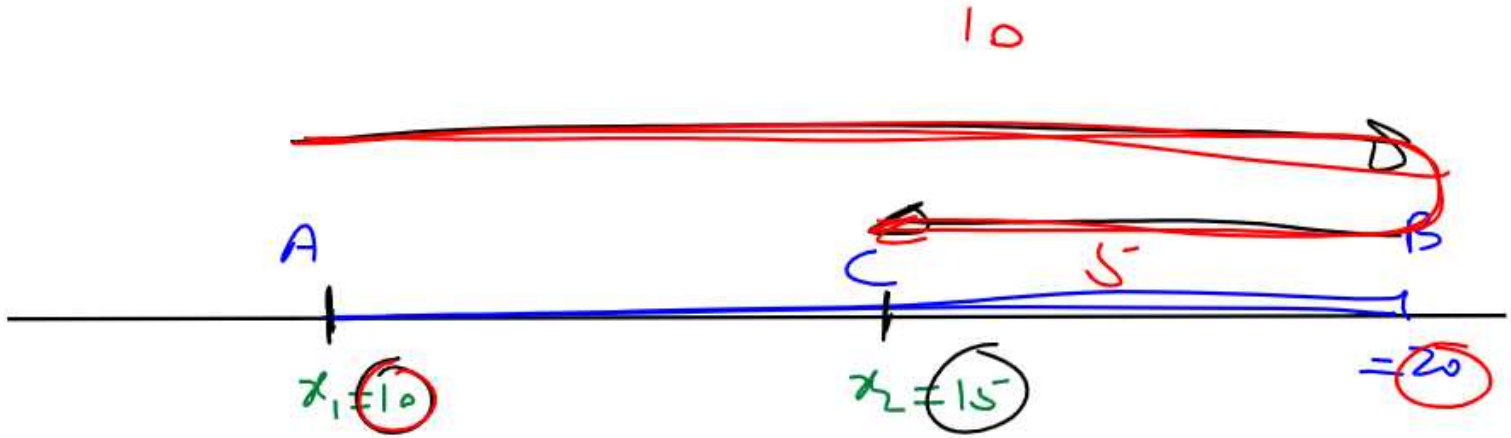
$= \Delta x$



$s = x_2 - x_1$

$= 15 - 10$

$= 5$



$\boxed{l = 15}$

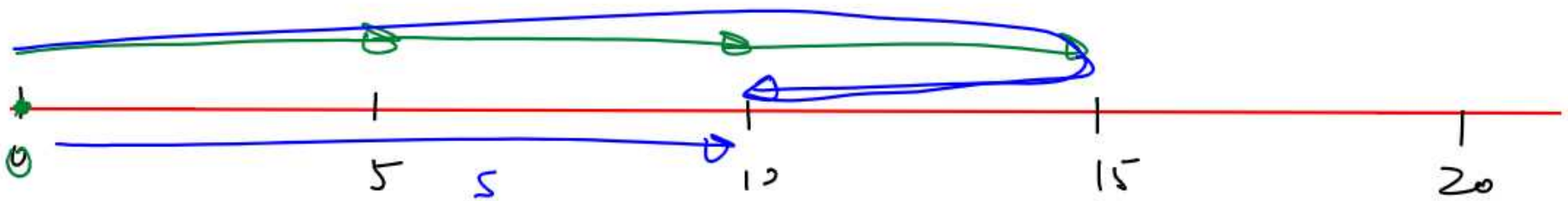
$l \geq |s|$

Distance (l)

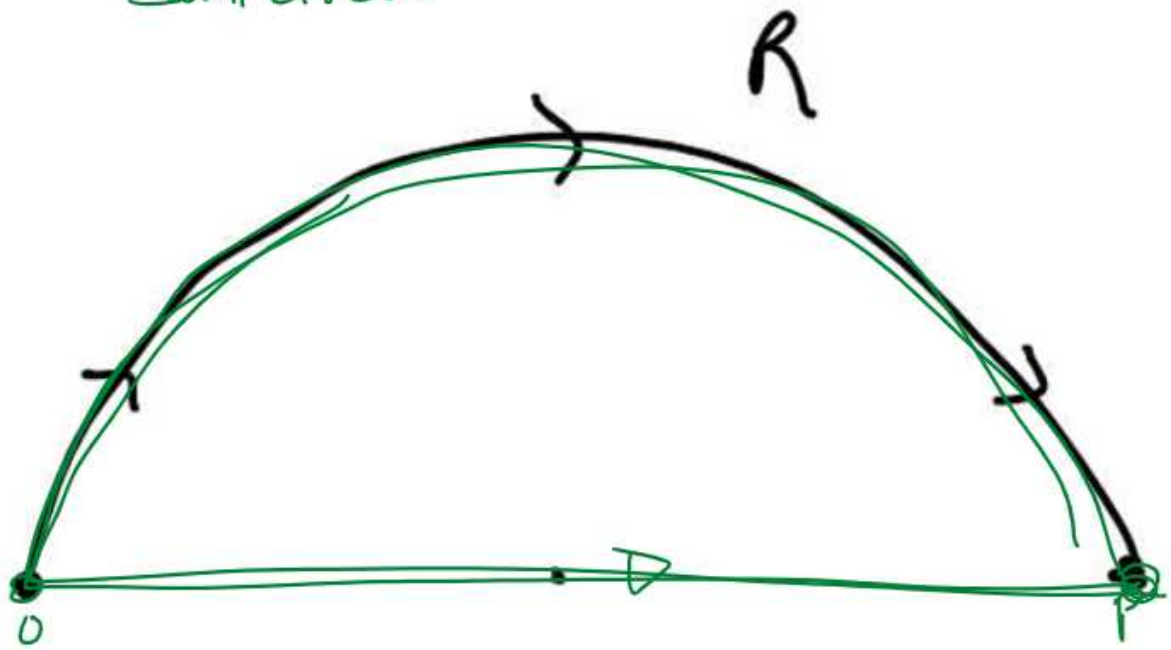
- Actual length of the path travelled is called distance.
- scalar quantity

- $l \geq |s|$

Equality will hold if direction remains unchanged


0
5
5
10
15
20

Semi Circle



Q. Find the distance & displacement of the mosquito

$l = \pi R$

$s = ?$

(A) 0

(B) πR

(C) $2R$

(D) R

Velocity (v)

$$\vec{v} = \frac{d\vec{r}}{dt} \text{ or } \frac{d\vec{s}}{dt}$$

- Velocity is rate of change of position with respect to time
- Vector quantity
- It can also be defined as rate of change of displacement wrt time

$$V = \frac{dx}{dt} \text{ or } \frac{ds}{dt}$$

- Avg. velocity $\frac{\text{displacement}}{\text{time}} \Rightarrow V_{avg} = \frac{s}{t}$

Speed (V_s)

- Rate of change of distance wrt time

$$V_s = \frac{dl}{dt}$$

- speed is magnitude of velocity

$$V_s = |v|$$

- Average speed $= \frac{\text{distance}}{\text{time}} \Rightarrow (V_s)_{avg} = \frac{l}{t}$

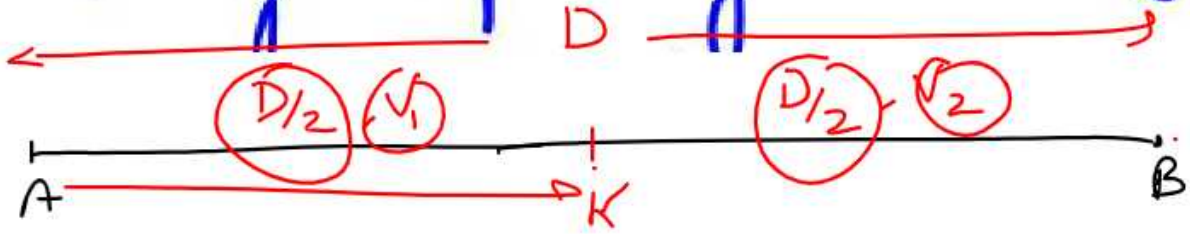
- $$(V_s)_{avg} \geq |v_{avg}|$$

$$\text{Avg. speed} = \frac{l}{t}$$

$$\text{Avg. velocity} = \frac{s}{t}$$

$$l \neq s$$

Q: A car moves at speed v_1 for first half of total distance and at v_2 for second half. Find average speed of entire journey.



(A) $v_1 + v_2$

(B) $\frac{v_1 + v_2}{2}$

(C) $\frac{2v_1v_2}{v_1 + v_2}$

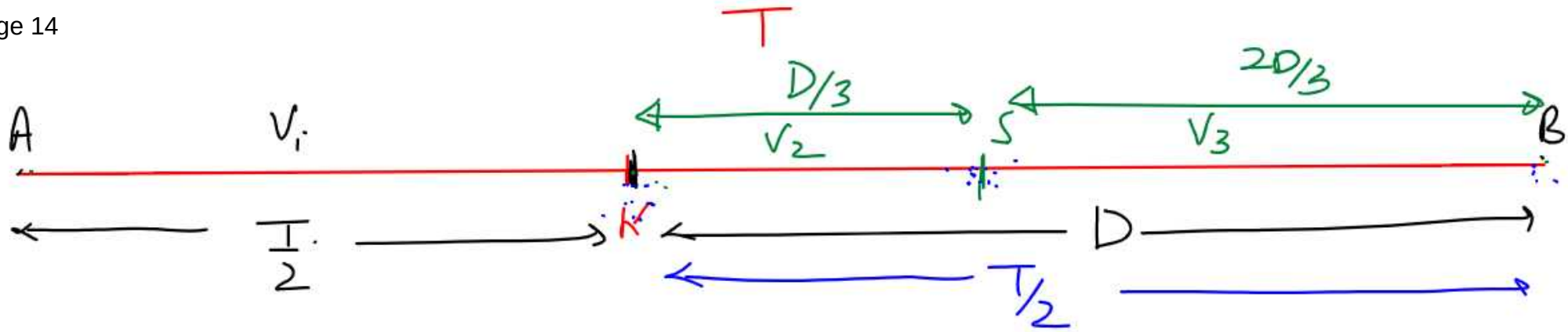
(D) $v_1 - v_2$

$$v_{avg} = \frac{D}{t_{AK} + t_{KB}}$$

$$= \frac{D}{\frac{D}{2v_1} + \frac{D}{2v_2}} = \frac{1}{\frac{1}{2v_1} + \frac{1}{2v_2}}$$

$$= \frac{2}{\frac{1}{v_1} + \frac{1}{v_2}} = \frac{2v_1v_2}{v_2 + v_1}$$

Q: A car travels at v_1 for first half of total time. For the remaining distance, first $\frac{1}{3}$ is travelled at v_2 and remaining at v_3. Find average speed of entire journey.



Avg. speed $= \frac{AB}{T}$

$= \frac{V_1 \frac{T}{2} + D}{T}$

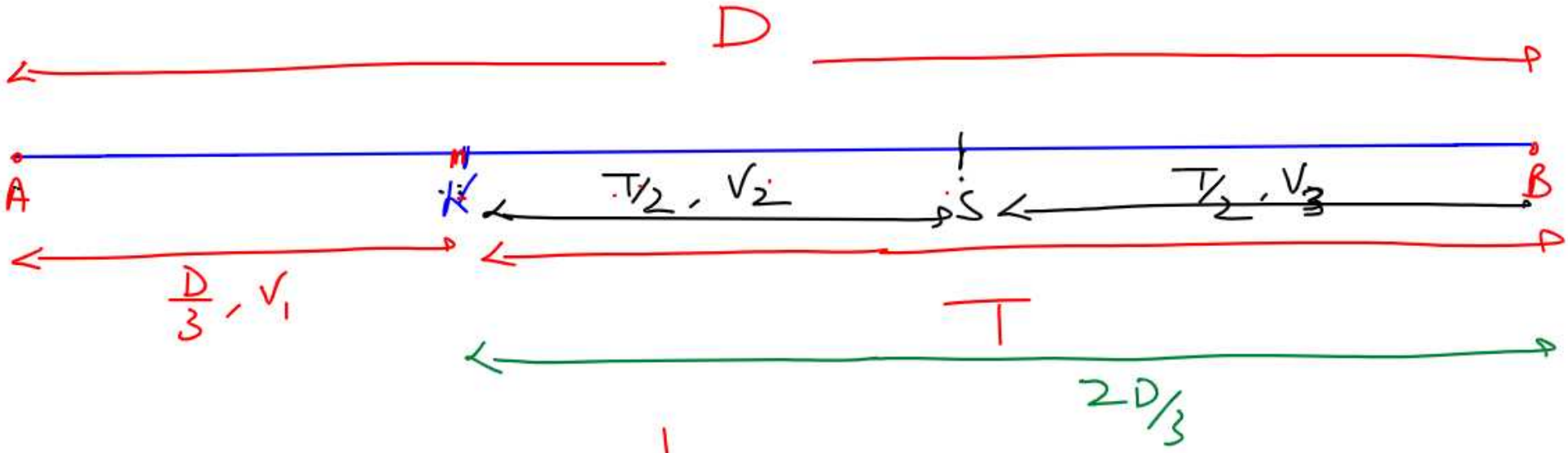
$\frac{D}{3V_2} + \frac{2D}{3V_3} = \frac{T}{2}$

$D = \frac{\frac{3T}{2}}{\frac{1}{V_2} + \frac{2}{V_3}}$

Avg. speed $= \frac{V_1 \frac{T}{2} + \frac{3T}{2}\left[\frac{1}{V_2} + \frac{2}{V_3}\right]}{T}$

$= \frac{V_1}{2} + \frac{3}{2}\left[\frac{1}{V_2} + \frac{2}{V_3}\right]$

Q. A car travels first $\frac{1}{3}$ of total distance at V_1. For the remaining time, half the time it travels at V_2 and remaining at V_3. Find V_{avg} of entire journey.



$$\text{Avg. speed} = \frac{D}{\frac{D}{3V_1} + T}$$

$$\frac{2D}{3} = KS + SB$$

$$\frac{2D}{3} = \frac{T}{2}V_2 + \frac{T}{2}V_3$$

$$T = \frac{4D}{3(V_2+V_3)}$$

$$\text{Avg. speed} = \frac{D}{\frac{D}{3V_1} + \frac{4D}{3(V_2+V_3)}}$$

$$= \frac{3}{\frac{1}{V_1} + \frac{4}{V_2+V_3}}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

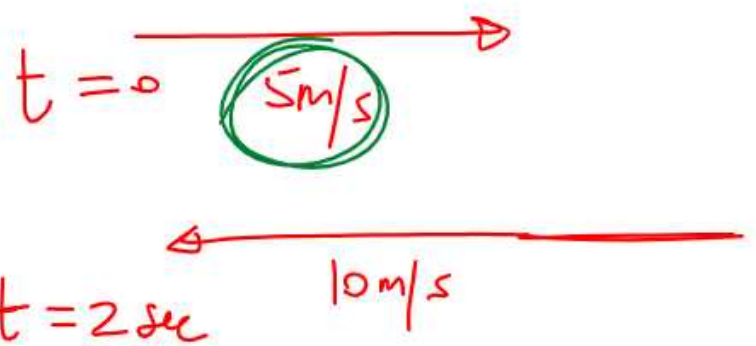
Acceleration (a)

- " is defined as rate of change of velocity wrt time

$$a = \frac{dv}{dt}$$

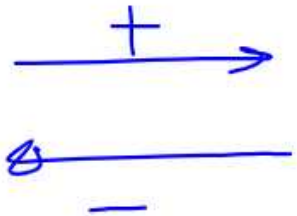
- Vector quantity
- $a_{avg} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$

Q. Will we say person is accelerated?



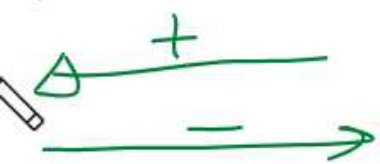
(A) Yes

(B) No



$$a_{avg} = \frac{V_2 - V_1}{t} = \frac{-10 - (5)}{2}$$

$$= \frac{-15}{2} = -7.5\,m/s^2 = 7.5\,m/s^2 \text{ (left)}$$

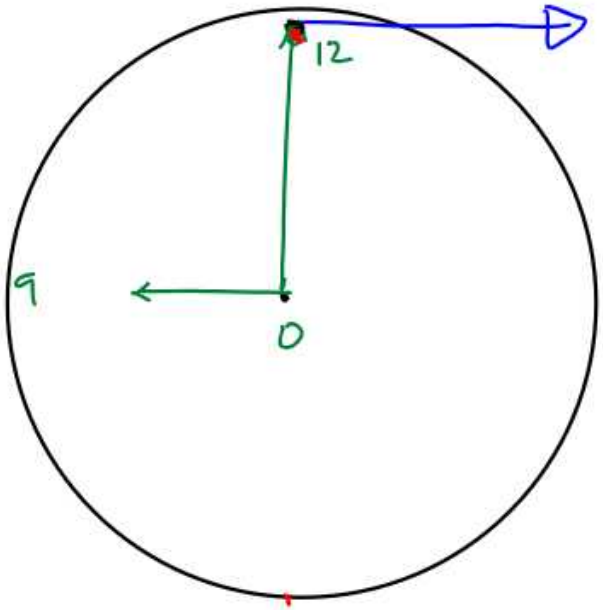


$$a_{avg} = \frac{V_2 - V_1}{t} = \frac{10 - (-5)}{2}$$

$$= \frac{15}{2} = 7.5\,m/s^2$$

$$= 7.5\,m/s^2 \text{ (left)}$$

Q: A clock is showing 9:00 hour time. Find the average velocity and average acceleration of the tip of the minute hand of the clock in next 30 mins if the length of the minute hand is 10cm.



Avg. Velocity $= \frac{20\text{cm}}{30\text{min}}$

$= \frac{2}{3}\text{ cm/min}$

$= \frac{2}{3\times 60}\text{ cm/sec}$

$= \frac{1}{90}\text{ cm/sec}$

Avg. Acceleration

$|a| = \frac{v_2 - v_1}{t_2 - t_1}$

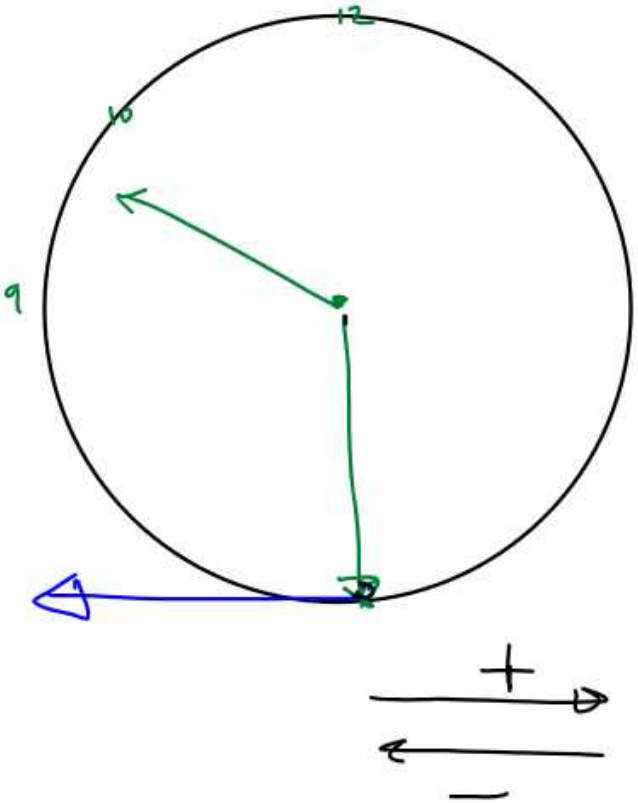
$v = \frac{2\pi \times 10\text{cm}}{60\text{ min}}$

$= \frac{\pi}{3}\text{ cm/min}$

$|a| = \frac{-\pi/3 - \pi/3}{30\text{min}}$

$= \frac{-2\pi}{90}\text{ cm/min}^2$

$|a| = \frac{2\pi}{90}\text{ cm/min}^2$ (left)



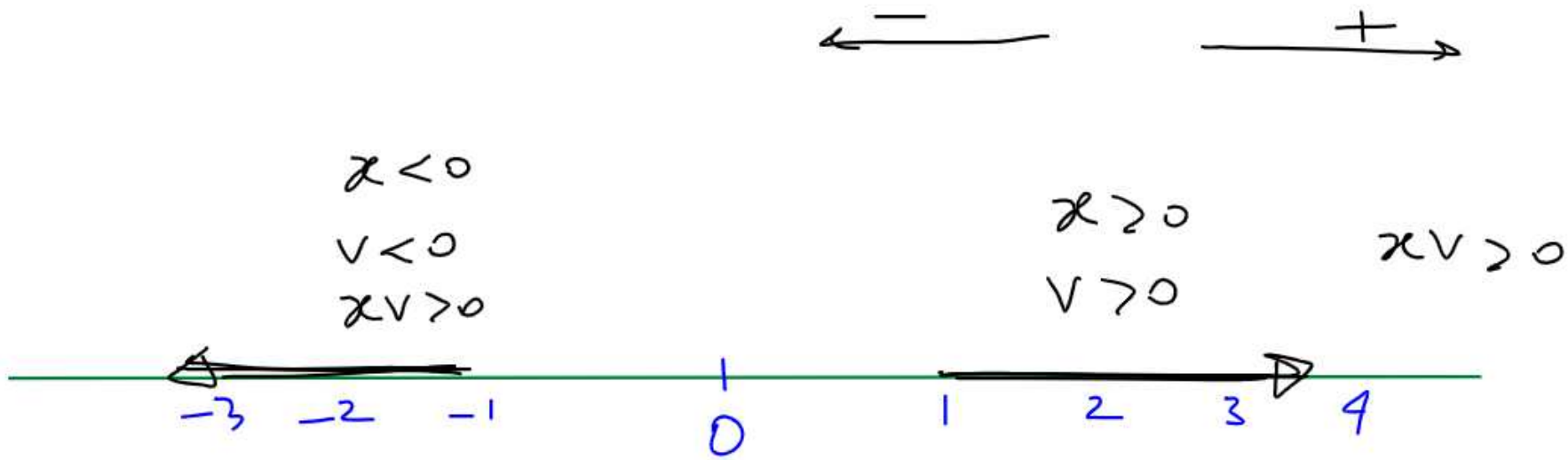
Q. Which of the following if true would mean that particle is going away from origin. (1-D Motion)

(A) $a \cdot v > 0$

(B) $x \cdot a > 0$

(C) $x \cdot v > 0$ ✓

(D) $v^2 > 0$



M1:

$x^2 \rightarrow$ increases

$\frac{d}{dt}(x^2) > 0$

$2x\frac{dx}{dt} > 0$

$xv > 0$

M2:

$xv > 0$

Three Equation of uniform motion ($a = \text{const}$)

(1) $v = u + at$

(2) $s = ut + \frac{1}{2}at^2$

(3) $v^2 = u^2 + 2as$

(4) $S_{nth} = u + \frac{a}{2}(2n-1)$

$$a = \frac{dv}{dt}$$

$$\int_u^v dv = \int_0^t a\,dt$$

$$v\Big|_u^v = a\int_0^t dt$$

$$v - u = a\,t\Big|_0^t$$

$$v - u = a\left[t - 0\right]$$

$$v = u + at$$

$$v = \frac{ds}{dt}$$

$$\frac{ds}{dt} = u + at$$

$$\int_0^s ds = \int_0^t u\,dt + \int_0^t at\,dt$$

$$s\Big|_0^s = ut\Big|_0^t + a\frac{t^2}{2}\Big|_0^t$$

$$s - 0 = u\left[t - 0\right] + \frac{a}{2}\left[t^2 - 0^2\right]$$

$$s = ut + \frac{1}{2}at^2$$

$$a = \frac{dv}{dt} = \frac{dv}{ds}\cdot\frac{ds}{dt} = v\frac{dv}{ds}$$

$$a\,ds = v\,dv$$

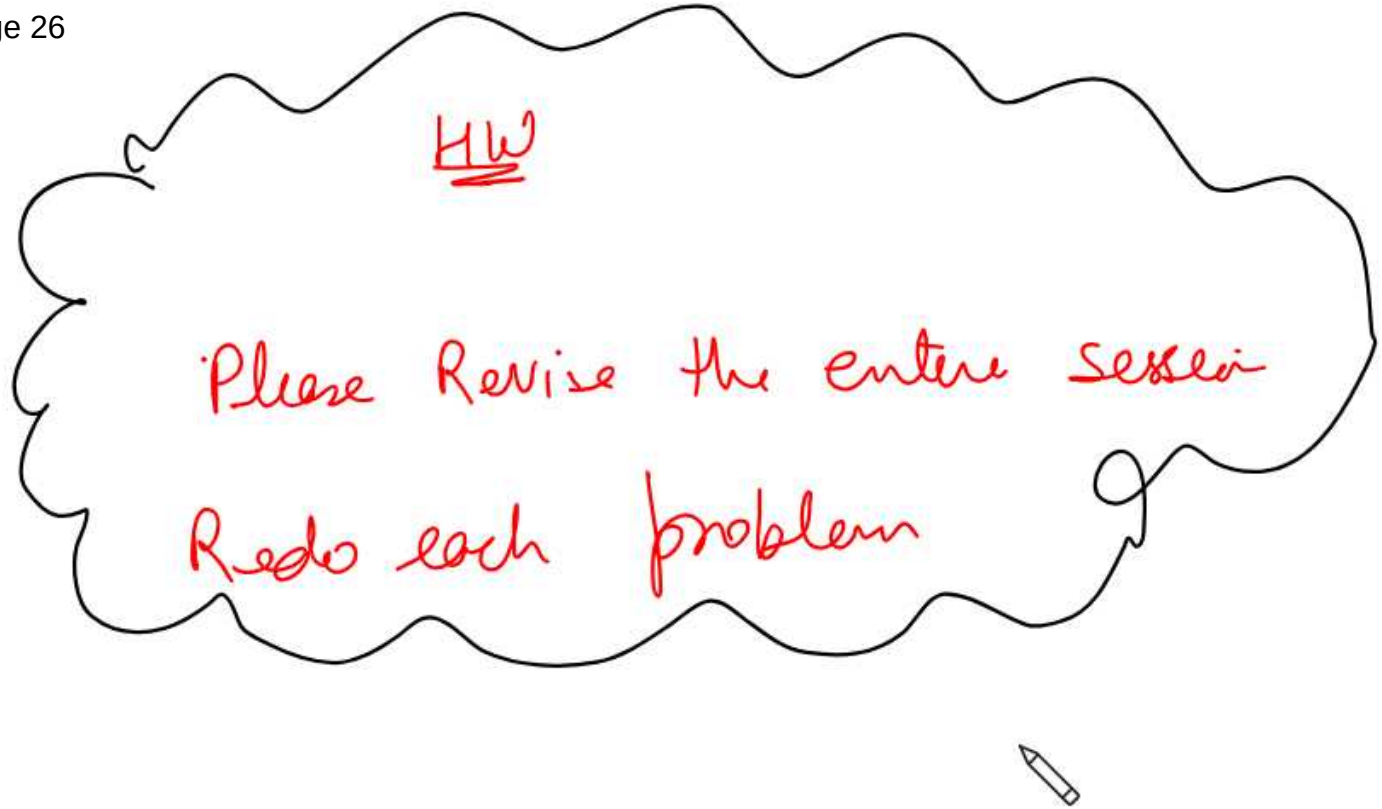
$$\int_0^s a\,ds = \int_u^v v\,dv$$

$$as\Big|_0^s = \frac{v^2}{2}\Big|_u^v$$

$$a[s-0] = \frac{1}{2}[v^2 - u^2]$$

$$2as = v^2 - u^2$$

$$v^2 = u^2 + 2as$$


HW
Please Revise the entire session
Redo each problem